

CMPE630: Digital IC Design Lab Report Rubric

The following provides a brief description of how grades are distributed among different components of the lab report. Keep in mind that **this document is merely a brief description of the report requirement**. Please check the lab report section of the lab handout, if applicable, for the specific report requirements.

Grading:

- For labs that require a report, **signoff is 60%** of the whole lab grade and the **report is 40%**
 - see the following pages for the report points breakdown
- See syllabus for late policy

Report Structure:

Your lab report will have the following:

- Cover Page
- Table of contents
- Abstract
- Design Methodology
- Result and Analysis
- Conclusions
- Questions
- References (if needed)
- Appendix (if needed/required)
- Signoff sheet

The details of the contents in each section can be found in the lab handout under the **Report Requirements** section. **This section is important as the TAs and the instructor will be looking there to see if you have everything included.** It is a unique section to each lab so be sure to look there each time before writing and submitting your report.

The remainder of this document will provide a general overview of each section and the corresponding grading rubric. **All lab reports will be graded out of 40 points.**

Cover Page (0.5pt)

- This should include the title of the lab, course number, your name, and the date on which the lab was performed

Table of Content (0.5pt)

- A table containing the title of each section in your report

Abstract (5pts)

- This should briefly state in your own words what you are trying to accomplish and why you are performing the lab. Do not simply copy or paraphrase the lab handout
- Please see the last page for an example Abstract
- ***DO NOT write a question. DO NOT re-write the lab handout. Give a synopsis of what you did, why you did it, and the major principles you employed to do it.***
- ***NO first person!***

Design Methodology (10pts)

- This section should contain any commentary related to:
 - Important theoretical components of the design
 - The motivation behind the design
 - Your transistor design and any equations or analysis performed to inform this design
 - Justification for design decisions
 - Symbols, Test Schematics, etc.
- ***DO NOT re-write the lab handout. DO NOT write a list of steps. In a technical report, you summarize what you did. You may use present or past tense, but be consistent. Give enough detail that a competent person could re-do exactly what you did and obtain the same results. Make sure you mention all components used and how they were used***
- ***Introduce and discuss all tables and figures within text***

Results and Analysis (10pts)

- This section will generally contain plots and other numerical results
- Include all relevant information gathered from running tests, building CMOS layout, etc.
- Explain **what** the results are and **why**
- ***Introduce and discuss all tables and figures within the text***

Conclusion (5pts)

- Restate important results, how they were obtained, and any uncertainties
- Explain the implications of your results

- Are there any possible future improvements? Is there anything you would do differently next time?
- Please see the last page for an example Conclusion

Questions (5pts)

- You will answer the questions in the lab manual here. Treat these like homework problems.

References

- If you used information/graphics from external sources (e.g. figures from lecture slides, or any figures that are not made by you), be sure to cite these references using the appropriate bibliographic style
 - Please use the IEEE format

Appendix

- This section can be used to include things that are too big for the report or things that are not crucial but may be helpful to the readers as references

The remaining (4pts) account for style, grammar, spelling, and formatting

- Note that these are not formal sections, but the aspects listed below will be evaluated throughout the lab report in its entirety
 - Figure and Table Labels (Axis, Titles, etc.)
 - Grammatical and Typographical Errors
 - Overall Report Quality
 - Font and Formatting Consistency

Please keep in mind that the information above is a generalized description of each section in your report. Please double-check the lab report requirement section of each lab to ensure that you have all of the required items in your report

General tips for improving report quality

- Reports should be written in the third person
- Any figure or table that appears in the report should be numbered and referred to in the text of the report
- Font style and size should remain constant throughout the report
 - There are no restrictions on font style/size. However, **Arial, 11pt font** is what I use
- Make sure that all figures and tables are large enough to be read by a reasonable person or even someone with poor eyesight (like me)
- Label each figure and table with a number and caption
- Strive to include figures that support claims, provide interesting results, or convey some new information regarding your lab

How much should I write?

There is **no required length** to the lab reports, but please keep the following in mind when you write your report:

- It is expected that students **show as much detail as they can, without being repetitive.** The goal of the lab reports is not only to practice formal documentation of work but also to ensure the students know the material that was covered in the lab.

Examples:

Below is an example Abstract and Conclusion from lab reports submitted by prior students in the course. I encourage you to use these as an example of a successful way to write these sections in your report. **However, this is not a secret formula that must be carefully followed, they are simply examples submitted by previous students that received high marks.**

Example Abstract:

CMOS inverters are arguably the most fundamentally important circuits for the design of digital integrated circuits. In this lab, we designed, simulated, and analyzed a static CMOS inverter using Mentor Graphics EDA tools and the TSMC 0.18 μm PDK. Individual MOSFETs were simulated to characterize their voltage transfer characteristics and threshold voltage. These data were used to guide the design and analysis of the CMOS inverter circuit, which was simulated under varying supply, load, and temperature conditions. The worst-case maximum input frequency and throughput for typical load conditions were estimated as XXX and XXX. Furthermore, the peak dynamic and the static power for worst-case conditions were measured as XXX and XXX, respectively.

Example Conclusions:

In this lab, we have designed and simulated a static CMOS inverter in TSMC 0.18 μm technology. Mentor Graphics EDA tools, including Pyxis Schematic and Eldo were used to capture the inverter's schematic and simulate its timing, as well as power characteristics. It was observed that the addition of a capacitive load reduces the overall performance of the inverter (maximum input frequency and throughput). This is a result of the additional charge that needs to be supplied to or drained from the inverter's output node to reach one of the supply rail voltages. The peak dynamic power (XXX), which was found to occur approximately halfway through the inverter's transition between low-to-high or high-to-low values, corresponds to a temporary short-circuit condition, where the NMOS and PMOS transistors short the supply rails. In addition, the static power consumption (XXX) creates a challenge for integrating the inverter into larger designs, especially in energy-constrained applications.